The experiment and the results are shown in figure 3. It could be seen that at generation zero, a single band at the bottom of the tube was produced. This is due to the fact that all the nitrogen in the DNA were the heavy ^{15}N isotope. At generation one, a single band was produced but in the intermediary position between where ^{15}N and ^{14}N isotopes were expected. This directly proves semi conservative replication as one strand is made of lighter isotope of nitrogen and the other strand is made of heavy isotope of nitrogen. Similarly the DNA at generation two showed two bands one at the intermediary region and another band at the lighter ^{14}N region. Subsequent generations of DNA showed these two bands, however the intermediary band was shown to become thinner and the ^{14}N band becoming broader. This is due to the fact that the strands containing ^{14}N isotope increases as the DNA replicates in ^{14}N medium.

6.2.6 Overview of DNA Replication

1. DNA replication is the process by which DNA makes a copy of itself during cell division.
2. The first step in DNA replication is to denature or 'unzip' the double helix structure of the DNA molecule, where the two strands of DNA are seperated.
3. DNA denaturation is carried out by an enzyme called helicase which breaks the hydrogen bonds holding the complementary bases of DNA together. Single strand binding (SSB) proteins bind to these strands to prevent them from annealing together.
4. Since the DNA is a very large molecule, the separation of the two single strands of DNA creates a 'Y' shape called a replication 'fork'.
5. The two separated strands will act as templates for synthesis of the new strands of DNA.
6. One of the strands is oriented in the 3' to 5' direction (towards the replication fork), this is the leading strand.
7. The other strand is oriented in the 5' to 3' direction (away from the replication fork), this is the lagging strand. As a result of their different orientations, the two strands are replicated differently.
8. Leading Strand replication: A short piece of RNA called a primer (produced by an enzyme called primase) comes along and binds to the end of the leading strand. The primer acts as the starting point for DNA synthesis. An enzyme called DNA polymerase binds to the leading strand and then 'walks' along it, adding new complementary nucleotide bases (A, C, G and T) to the strand of DNA in the 5' to 3' direction in a continuous fashion.
9. Lagging strand replication: Numerous RNA primers are made by the primase enzyme and bind at various points along the lagging strand. Chunks of DNA, called Okazaki fragments, are then added to the lagging strand also in the 5′ to 3′ direction, by DNA polymerase. This type of replication is called discontinuous as the Okazaki fragments will need to be joined up later.
10. Once all of the bases are matched up (A with T, C with G), an enzyme called exonuclease strips away the primer(s). The gaps where the primer(s) were are then filled by yet more complementary nucleotides.

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